

(CCNA 200-301)

Ser No	Nomenclature	Items and their Technical Qualitative Requirement	Remarks
1.	Network Fundamentals :-		
	Role and function of network components	OSI & TCP/IP Model	
		Routers	
		Layer 2 and Layer 3 switches	
		Next-generation firewalls and IPS	
		Access points	
		Controllers (Cisco DNA Center and WLC)	
		Endpoints	
		Servers	
		PoE	
	Network Topologies	Star	
		Mesh	
		Hybrid	
		Two-tier	
		Three-tier	
		Spine-leaf	
		WAN	
	Physical interface and cabling types	• Single-mode fiber • multimode fiber • copper	
		Connections (Ethernet shared media and point-to-point)	
		Interface and cable issues	
		Collisions, errors, mismatch duplex, and/or speed	
	Wireless Principles	Non overlapping Wi-Fi channels	
		SSID	
		RF	
		Encryption	
	IP Addressing	Sub netting	
		FLSM,	
		VLSM	
	IPv4 Address	Classes	
		Unicast	
		Broadcast	
		Multicast	
	IPv6 addressing and prefix and IPv6 address types	Unicast (global, unique local, and link local)	
		Any cast	
		Multicast	
		Modified EUI 64	
		Auto-configuration	

Ser No	Nomenclature	Items and their Technical Qualitative Requirement	Remarks
2.	Network Access :-		
	Switching concept	MAC learning and aging	
		Frame switching	
		Frame flooding	
		MAC address table	
		Layer 2 & Layer 3 Switch Operation	
		Ethernet frame format	
	VLAN	Access ports	
		Default Vlan	
		Private Vlan	
		Intervlan connectivity	
	Inter switch connectivity	Trunk ports	
		Add and remove Vlan on trunk	
		VTP & 802.1Q	
		Native VLAN	
		SPAN	
		SNMP	
		IP SLA	
	Layer 2 discovery protocol	Cisco Discovery protocol	
		LLDP	
	Layer 2/Layer 3 Ether Channel	Static	
		PAGP	
		LACP	
	Switch Port Security	Static	
		Dynamic	
		Sticky	
		Max MAC addresses	
		Violation actions	
		Err-disable recovery	
	Spanning Tree Protocol	PVST+	
		RPVST+	
		Root port, root bridge (primary/secondary), and other port names	
		STP root bridge selection	
		PortFast	
		Port states	
		BPDU guard	
		MSTP	
		Cisco Wireless architecture and Access point modes	
	Physical infrastructure connections of WLAN components	Access point	
		WLC	
		Access/trunk ports	
		LAG	

Ser No	Nomenclature	Items and their Technical Qualitative Requirement	Remarks
	First Hop Redundancy Protocol	HSRP	
		VRRP	
		GLBP	
		Voice Vlan and Wireless Vlan	
	Components of routing table	Routing protocol code	
		Prefix	
		Network mask	
		Next hop	
		Administrative distance	
		Metric	
		Gateway of last resort	
3.	IP Connectivity :-		
	Static & Default Routing	Default route	
		Network route	
		Host route	
		Floating static	
		Configuration & Troubleshooting	
	RIP Routing Protocol	RIPv1	
		RIPv2	
		RIPng	
		Configuration & Troubleshoot	
	Enhanced Interior Gateway Routing Protocol	IPv4 / IPv6	
		Authentication	
		Summarization	
		Redistribution	
		Stub	
		Passive Interface	
	Open Shortest Path First (OSPFv2 / OSPFv3)	IPv4/IPv6	
		Neighbour adjacencies	
		Point-to-point	
		Broadcast (DR/BDR selection)	
		Router ID	
		Virtual link	
		LSAs	
		Filtering	
		Summarization	
	Border Gateway Protocol	IPv4/IPv6	
		IBGP	
		EBGP	
		Synchronization	
		Redistribution	
		Authentication	
		Redistribution In IPV4 Technology	

Ser No	Nomenclature	Items and their Technical Qualitative Requirement	Remarks
		Redistribution In IPV6 Technology	
		MD 5 Authentication On All Routing Protocol	
4.	IP Services:-		
	Remote Access protocols	Telnet	
		SSH	
		Access-list, Prefix-list, Route-map, Filter-list, Distribution-list Configuration	
	Access-list	Standard access-list	
		Extended access-list	
		Named	
	Network Address Translation	Static NAT	
		Dynamic NAT	
		PAT overload	
	Network Time Protocol	Server mode	
		Client mode	
	Dynamic Host Control Protocol	DHCP server	
		DORA process	
		DHCP client	
		DHCP relay	
		DHCP snooping	
	Simple Network management Protocol	Function	
		Network operation	
	Domain name system	DNS servers	
		DNS pointers	
		DNS attacks	
		DNS lookup	
		FTP and TFTP	
	Syslog	Features	
		Levels	
		Uses	
		Packet Capturing & Monitoring On Wireshark	
5.	WAN Technologies :-		
	Wide Area Network topologies	Point-to-point	
Hub and spoke			
Full mesh			
Single and Dual-homed			
PPP and MLPPP on WAN interface using local authentication			
PPPoE client-side authentication			
	Quality Of	classification	

Ser No	Nomenclature	Items and their Technical Qualitative Requirement	Remarks
	Service	marking	
		device trus	
		queuing	
		Prioritization • Voice • Video • Data • Shaping • Policing • congestion management	
		Metro Ethernet	
		Single homed branch connectivity using eBGP IPv4	
	Multi-Protocol label Switching	CEF	
		LDP	
		PUSH/POP/SWAP	
		VRF	
		MBGP	
6.	Security Fundamentals :-		
	Security concepts	threats	
		vulnerabilities	
		exploits	
		mitigation technique	
		Device access control using local password	
	Security password policy elements	multifactor authentication	
		Certificates	
		biometrics	
		Authentication, authorization and accounting (AAA)	
		Device security using RADIUS and TACACS	
	Wireless security protocol	WPA	
		WPA2	
		WPA3	
		WLAN using WPA2 PSK	
		Layer 2 ARP Inspection	
		GRE tunnel VPN	
		DM-VPN	
		IP-sec remote access VPN	
		SSL VPN	
		L2TP VPN	
		PPTP VPN	
		MPLS VPN	
		Configuring and troubleshooting on Cisco ASA	
		Configuring and troubleshooting on Sophos firewall	
		Configuring and troubleshooting on Fortinet firewall	

